A Real-Time Unsupervised Hyperspectral Band Selection via Spatial–Spectral Information Fusion-Based Downscaled Region

Chenglong Zhang, Lichao Mou, Xiaoli Yang, Xiangrong Zheng, Xiao Xiang Zhu, *Fellow, IEEE*, and Xiaopeng Ma, *Member, IEEE*

***Abstract*— Information fusion plays a vital role in hyperspectral band selection as it enables the exploration of the spatial–spectral structure relationship present in bands of hyperspectral images (HSIs). However, the focus of most algorithms primarily lies in the processing of single-band vectors, leaving only a few algorithms to handle spatial–spectral features of HSIs in a complex and inefficient manner. To overcome these limitations, a real-time unsupervised hyperspectral band selection method via spatial–spectral information fusion-based downscaled region (SIFDR) is proposed in this study. In particular, this approach incorporates an energy constraint method for assigning band weights to each detected pixel and estimates band spectral information through average fusion. Furthermore, a band weak redundancy sorting method is introduced, which is based on spectral information peaks, thereby achieving complementary spectral information. By performing regional downscaling of the HSI, spatial–spectral information is effectively fused, resulting in a real-time entire process. To evaluate the effectiveness of the proposed algorithm, experiments were conducted on four hyperspectral datasets, including an ultrahigh-dimensional medical HSIs, which distinguishes itself from previous methods that are typically evaluated exclusively on remote sensing datasets. Comparative results with several state-of-the-art (SOTA) algorithms demonstrate that the proposed algorithm excellently accomplishes hyperspectral band selection tasks in real time. The code of SIFDR has been shared on https://github.com/zhangchenglong1116/SIFDR.**

***Index Terms*— Dimensionality reduction, hyperspectral band, information fusion, real-time.**

NOMENCLATURE

$X \in \mathbb{R}^{l\times n}$	Hypercube of the raw hyperspectral images (HSIs).
l	Number of total bands for X.
n	Number of total pixels for X.
$x_i \in \mathbb{R}^{l\times 1}$	Vector of the ith pixel for X.
$R \in \mathbb{R}^{l\times l}$	Autocorrelation matrix of X.
$d \in \mathbb{R}^{l\times 1}$	Target a priori spectral vector.
$w \in \mathbb{R}^{l\times 1}$	Finite impulse response filter.
$W \in \mathbb{R}^{l\times n}$	Weight matrix for X.
$\rho \in \mathbb{R}^{l\times 1}$	Band weight vector for X.
D_{ij}	Euclidean distance (ED) between the ith and jth band.
$D \in \mathbb{R}^{l\times l}$	ED matrix between bands corresponding to X.
$\delta \in \mathbb{R}^{l\times 1}$	Band measures reflect redundancy.
$\varphi \in \mathbb{R}^{l\times 1}$	Comprehensive ranking metrics for bands.
$\check{X} \in \mathbb{R}^{l\times m}$	Downscaled image of X.
m	Number of total pixels for $\check{X}$.
r	Spatial scaling factor for bicubic interpolation.
t	Number of selected bands.
$Y \in \mathbb{R}^{t\times n}$	Selected hypercube.

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I. INTRODUCTION

HYPERSPECTRAL imaging possesses the capability to capture the continuous spectral information of reflections or transmissions within a specific range of wavelengths, resulting in the formation of numerous bands with abundant spectral and spatial details [1]. In comparison to conventional grayscale or RGB images, hyperspectral images (HSIs) offer a more discerning physical foundation [2]. As a result, they find extensive applications in various fields such as agriculture [3], environmental monitoring [4], mineral exploration [5], food safety [6], and medical diagnosis [7]. The advancement of deep learning models for HSI classification has further accelerated these applications [8], [9], [10]. However, the high-dimension of hyperspectral data prevents accurate classification with limited training samples, leading to the "curse of dimensionality." Moreover, this high dimension imposes significant computational burdens on data processing and analysis. Consequently, addressing the reduction of dimensionality in HSIs is imperative.

Dimensionality reduction of HSIs can be achieved through either feature extraction or feature selection, also known as band selection. Feature extraction involves projecting high-dimensional data into a lower dimensional space by implementing a specific embedding method to achieve

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dimensionality reduction [11]. Examples of common feature extraction methods include principal component analysis (PCA) [12], maximum noise separation (MNF) [13], and linear discriminant analysis (LDA) [14]. Although feature extraction reduces data dimensionality, it alters the data structure, resulting in a lack of physical meaning and potential loss of vital information. In contrast, the hyperspectral band selection strategy constructs a new image cube by selecting representative bands from the original data [15]. This approach compresses dimensions while preserving the physical meaning of HSIs, avoiding spectral information loss, and ensuring the interpretive reliability of HSIs. Consequently, hyperspectral band selection appears to be a more viable method for dimensionality reduction.

According to the allocation of labeled samples, band selection methods can be categorized into three types: supervised, semi-supervised, and unsupervised. Supervised and semi-supervised methods utilize labeled samples to train models, aiming to optimize the separation of samples and minimize the clustering of similar samples through the selection of bands [16]. With a sufficient number of samples, these methods can effectively identify the most discriminative spectral bands. However, gathering labeled samples is often challenging in real-world scenarios. On the other hand, unsupervised band selection methods do not require prior knowledge of labeled samples. Instead, they analyze and model the statistical properties and structure of the image itself to achieve band selection. For instance, a variety of indicators, such as variance [17], information entropy [18], and local density [19], are developed to evaluate the importance of bands. As a result, this study concentrates on the development of an unsupervised band selection algorithm. In the field of unsupervised band selection, clustering and ranking are the two primary strategies employed. Clustering-based methods involve clustering all bands and then selecting representative bands from each cluster, such as choosing the band at the center of the cluster [20] or the band with the highest information entropy [21]. However, these methods face limitations due to the performance of clustering algorithms, as noise bands often act as outliers that corrupt the clustering results. Conversely, sorting-based algorithms offer a simple and efficient approach [22]. Their core idea revolves around constructing an importance evaluation index for each band based on the statistical characteristics of the data, which is then utilized to rank all bands. A subset of bands can subsequently be formed by choosing the higher ranked ones. However, this approach typically only considers the correlation between individual bands, failing to adequately capture the intricate relationships between multiple bands and potentially disregarding the combination of important features. Additionally, search algorithms that employ intelligent optimization algorithms to iteratively find the optimal band subset have also attracted attention [23], [24]. Nonetheless, this type of algorithm often faces the challenge of a cumbersome and inefficient iteration process. More recently, scholars have begun to explore deep learning band selection techniques with nonlinear feature representation capabilities. These techniques primarily automate the generation of advantageous features for band clustering or ranking. It is worth noting that compared to spectral-dimensional convolution methods that solely utilize the statistical properties of HSIs, convolutional neural networks (CNNs) [25], and graph convolutional networks (GCNs) [26], which focus on spatial structure information of the data, can yield more stable results. This highlights the effectiveness of fusing spatial–spectral information to represent the structural correlations among various bands. However, this method significantly increases the inference time of the deep learning algorithm and is not suitable for real-time applications. While region segmentation can improve spatial–spectral fusion efficiency [27], the presence of noisy bands often leads to the incorrect generation of superpixels, thereby impairing band selecting performance.

In the process of band selection, researchers encounter two principal challenges: First, developing an effective metric that balances information and redundancy within the bands is formidable. Second, the integration of spatial and spectral information, although crucial, is time-consuming and impedes real-time processing. These issues are particularly acute in HSIs that contain noisy bands. Consequently, it is essential to develop a framework that supports efficient and real-time band selection through spatial–spectral information fusion.

To this end, a real-time unsupervised hyperspectral band selection framework via spatial–spectral information fusion based downscaled region (SIFDR) is proposed in this study. First, we draw inspiration from the constrained energy minimization (CEM) algorithm and develop a new evaluation metric that summarizes the band contributions of each detected pixel to estimate the spectral information of each band. Additionally, a strategy that ranks bands with peak information is employed to avoid redundancy and enhance the complementarity of spectral information. To mitigate the interference of noise bands, the hyperbolic tangent function is introduced to normalize band distances. Moreover, to integrate spatial and spectral information, a fusion method based on downscaled region is adopted. Specifically, the bicubic interpolation algorithm is utilized to compress the original image spatially. This enables the integration of spatial structure information into a smaller scale image, thereby laying the foundation for real-time processing.

The primary contributions of this article can be summarized as follows.

1) A succinct and efficient SIFDR band selection framework is presented in this study, which models the relationship between detected pixels and the entire frequency band via CEM. By utilizing the average band weight corresponding to all pixels, a more precise estimation of the spectral information for each band in the HSI can be achieved.
2) To enhance the complementarity of spectral information, a band ranking strategy based on information peaks has been implemented. The avoidance of redundancy is ensured by selectively choosing bands that exhibit diverse and significant spectral responses. Additionally, the effective noise suppression technique of hyperbolic tangent normalization has been creatively introduced.
3) A spatial–spectral fusion framework based on downscaled region is devised, employing bicubic interpolation

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to compress the original image space and incorporate spatial structure information on small-scale areas. This approach significantly decreases the computational burden, enabling real-time execution of the entire SIFDR process.

4) In contrast to the experimental design employed in previous studies, our research involved verification not only on traditional remote-sensing images but also on an ultrahigh-dimensional medical HSIs. Through a comparison with several state-of-the-art (SOTA) band selection algorithms, SIFDR demonstrates exceptionally high performance.

The structure of this article consists of several sections. Section II introduces different unsupervised hyperspectral band selection methods, including clustering and ranking approaches. Section III presents the detailed proposed SIFDR framework, along with an analysis of its computational complexity. Section IV describes the validation results and analysis carried out on four public datasets. Finally, Section V concludes the study.

II. Related Work

In this section, we review several classic and contemporary algorithms in the field of band selection. These algorithms are divided into two main groups, clustering-based and ranking-based, to provide an introduction.

A. Clustering-Based Algorithms

The band selection method, based on clustering, aims to cluster all available bands using an algorithm. Subsequently, it selects representative bands from each cluster to form a new subset [28]. Various strategies for grouping bands have been proposed in prior studies. Wang et al. [29] introduced the adaptive subspace partition strategy (ASPS) for band clustering. This approach focuses on maximizing the ratio of interclass distance to intraclass distance and selecting the least noisy band. They also proposed the fast neighborhood grouping method for band selection (FNGBS) [30], which employs a coarse-fine strategy to divide the hyperspectral cube into multiple groups. The band with the highest information entropy and local density is then chosen from each group. Additionally, hierarchy-based clustering structures, such as adaptive distance-based band hierarchy (ADBH) and hyperbolic clustering-based band hierarchy (HCBH) [31], [32], are widely used for band selection. Moreover, using deep learning to extract deep features for band clustering is a novel means. For instance, the recently proposed iterative graph autoencoder for band selection (IGAEBS) exhibits better clustering performance compared to previous methods [33]. Furthermore, other methods like subspace clustering and tensor learning are developed [34], [35].

Spatial information is instrumental in revealing the distribution of spectral bands within high-dimensional space. As a result, a practical approach involves integrating spatial and spectral information. Typically, this is accomplished by extracting features from single or multiscale spatial blocks and subsequently employing spectral clustering techniques. However, this strategy can be time-consuming. To mitigate computational demands, spatial–spectral fusion can be achieved by utilizing superpixels obtained through segmentation algorithms, and superpixels serve to represent homogeneous regions [36]. This approach has shown effectiveness in scenarios where the spectral bands are relatively pure, such as region-aware latent features fusion based clustering (RLFFC) and hierarchical latent feature representation learning-guided clustering (HLFC) [27], [37]. However, obtaining accurate superpixels from raw data containing noisy bands can be challenging, consequently limiting the algorithm's performance. The proposed SIFDR framework successfully achieves a harmonious combination of efficiency and effectiveness, allowing for real-time and precise processing.

B. Ranking-Based Algorithms

Ranking-based band selection ranks bands based on a metric to create a subset composed of the top-ranked bands. The original approach only considered the individual statistics of each band, such as information entropy, variance, and signal-to-noise ratio (SNR). Guo et al. [18] proposed Band selection using mutual information (MI). Maximum variance PCA (MVPCA) and constrained band selection (CBS) is documented in [17] and [38]. However, these statistics fail to account for the correlation between bands, resulting in redundant subsets. To address this, the enhanced fast density-peak-based clustering (EFDPC) algorithm is proposed, which ranks bands by quantifying their potential to become cluster centers [20]. This approach considers both the relevance and importance of the bands. Similarly, Sun et al. [39] proposed the example component analysis (ECA), which shares a similar concept. In addition, the band weights optimized by deep learning as priorities is also a widely utilized strategy. An example of this is the recently proposed novel stochastic gate-based autoencoder (SGAE) enhances the reliability of selected bands by constructing a nonlinear regularization loss [39]. Furthermore, a series of improvement methods have been introduced, such as those based on fully connected networks [40], CNNs [41], and graph CNNs [26]. However, these methods rely on the reconstruction error with the original image as the primary loss function, which is not suitable for HSIs that contain noisy bands. Moreover, the complex iterative process and numerous hyperparameter settings hinder real-time implementation.

C. Searching-Based Algorithms

These algorithms frame band selection as an optimization problem, aiming to identify the optimal subset of bands from high-dimensional spectral data. Initial research often combines intelligent optimization techniques with tailored objective functions to select bands that align closely with downstream supervised learning tasks. For instance, Zhang et al. [42] combined metrics of classification accuracy with the reciprocal of the number of bands, utilizing the gravitational search algorithm (GSA) to maximize classification accuracy with a minimal number of bands. Similarly, leveraging label information from training samples, Su et al. [43] used the particle

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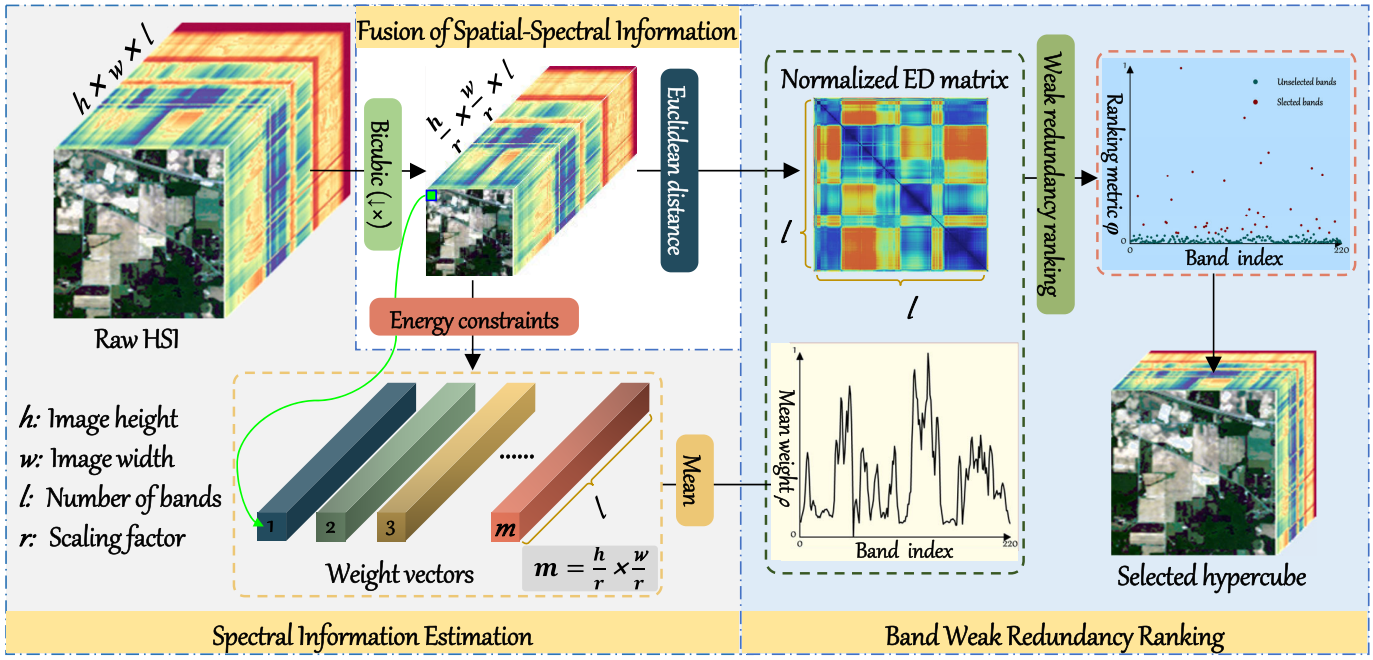


Fig. 1. Flowchart of SIFDR.

swarm optimization (PSO) algorithm to filter bands with maximum class separability, enhancing search efficiency. Contemporary studies increasingly adopt unsupervised methods to identify band subsets that maximum information minimum redundancy (MIMR). Feng et al. [44] utilized the clonal selection algorithm (CSA) to find bands with the highest MI, achieving MIMR. Furthermore, modified discrete GSA was designed to improve performance (MIMR-DGSA) [45]. Ou et al. [24] treated MIMR as a multiobjective optimization challenge, employing an advanced cuckoo search algorithm (MOCS), representing the SOTA in unsupervised search paradigms. Notably, the iterative search process can be time-consuming, and algorithm performance largely depends on precise parameter settings, such as population size, crossover rate, and mutation rate; inappropriate settings can significantly impair algorithm effectiveness.

III. Methodology

The SIFDR framework, as depicted in (Fig. 1), encompasses several key steps. Initially, the input raw HSIs undergoes a band-by-band downscaling procedure to integrate spatial–spectral information effectively. Subsequently, spectral information is estimated by employing energy constraints. Finally, a weak correlation-based sorting technique is applied to all bands, resulting in the selection of the top bands that constitute a novel hyperspectral cube. A detailed explanation of the algorithm will be provided next, the key notations employed in this article are summarized in Nomenclature.

A. Spectral Information Estimation

The abundant spectral information inherent in HSIs is crucial. While it can effectively differentiate target objects, not all spectral bands make significant contributions to their recognition. Band contributions mainly depend on whether they carry distinctive information. Thus, the key to selecting hyperspectral bands lies in the effective evaluation of the quality of spectral information. In hyperspectral target detection tasks, CEM constrains the output energy through spectral filters to suppress the background and highlight the target signal [46]. Inspired by this, we developed a spectral information estimation method based on energy constraints.

1) Constrained Energy Minimization: The CEM method aims to extract target signals from specific directions while suppressing interfering signals from other directions. It draws inspiration from linearly constrained minimum variance beamformers commonly used in signal processing. When the target a priori spectral vector d is known, to extract the target signal in a specific direction, a finite impulse response filter w is designed. The primary objective of this filter is to minimize the output response of the image pixel while ensuring that the response value corresponding to the target spectrum of interest is precisely 1. For a pixel x_i to be measured, its output response value after filtering is

$$y_i = x_i^T w. \tag{1}$$

Hence, the average energy output of all pixel responses within the image can be expressed as

$$\frac{1}{n}\sum_{i=1}^{n} y_i^2 = \frac{1}{n}\left[\sum \left(x_i^T w\right)^T \left(x_i^T w\right)\right] = w^T\left(\frac{1}{n}\sum x_i x_i^T\right) w = w^T R w \tag{2}$$

where $R = (1/n)\sum x_i x_i^T$ represents the autocorrelation matrix of HSIs $X \in \mathbb{R}^{l\times n}$.

In summary, the CEM method can be reduced to the optimization problem of determining the weight vector w of

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the filter

$$\underset{w}{\operatorname{argmin}}\left(\frac{1}{n}\sum_{i=1}^{n} y_i^2\right) = \underset{w}{\operatorname{argmin}}\left(\omega^T R\omega\right), \quad \text{s.t. } d^T w = 1. \tag{3}$$

By employing the Lagrange multiplier method, we can obtain the optimal value for w

$$w = \frac{R^{-1}d}{d^T R^{-1}d}. \tag{4}$$

In this way, w filter is performed on all pixels of the entire image to complete the detection of target d

$$y_i = x_i^T w = \frac{x_i^T R^{-1}d}{d^T R^{-1}d}. \tag{5}$$

Clearly, the fundamental principle of CEM for target detection essentially involves a weighted summation of various spectral bands. The absolute value of the weight vector $|w|$ reflects the importance of each band to the detection target d, bands with higher weights often contribute richer information. Consequently, this impels our pursuit to estimate the information of spectral bands by leveraging the utilization of CEM weights.

2) Spectral Information Estimation: In Section III-A1, we emphasized that the acquisition of the band weight vector solely relies on the detection of target d. Nonetheless, in practical full-image application tasks, distinct pixels often correspond to diverse target categories. Consequently, evaluating the band solely based on a weight vector obtained from a single target becomes challenging. To tackle this predicament, we adopted a comprehensive approach. Precisely, we treated every pixel in the image as an independent target, computed the corresponding CEM weight vector, and combined them to obtain the matrix $W = [w^1, w^2, \ldots, w^n] \in \mathbb{R}^{l\times n}$, subsequently, we took the mean of the absolute values of each row in W achieved by l_1 -norm form to obtain, which utilized for estimating information. The calculation details are outlined as follows:

$$W = \frac{R^{-1}X}{X^T R^{-1}X}, \tag{6}$$

$$\rho = \frac{1}{n}\{\|W(i,:)\|_1\}_{i=1}^{l}. \tag{7}$$

In general, the number of pixels, n, tends to be quite substantial. Consequently, to effectively manage the computational resources while maintaining a balance between memory utilization and computational speed, we adopt a batch calculation approach for determining W after obtaining R. In this study, we have specifically chosen a batch size of 1000 pixels per group. The pixels of the ith batch are

$$X^i = \left[x_{1000\times(i-1)+1}, \ldots, x_{\min(1000\times i,\, n)}\right]. \tag{8}$$

It is important to highlight that, to mitigate the significant disparities in weights resulting from the diverse magnitudes of different bands, we employ a normalization process for each band. Before computing W, we normalize the bands using the following formula:

$$X = \left\{\frac{X(i,:) - \min(X(i,:))}{\max(X(i,:)) - \min(X(i,:))}\right\}_{i=1}^{l}. \tag{9}$$

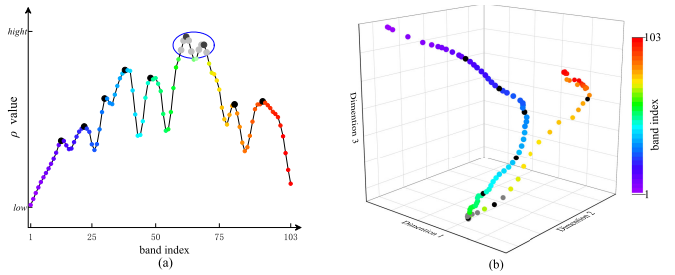


Fig. 2. (a) ρ value associated with the bands in Pavia University was computed. (b) Spatial distribution of the bands was visualized after dimensionality reduction using PCA.

B. Band Weak Redundancy Ranking

Although we have introduced a new indicator, ρ, to measure the information content of spectral bands, it is not advisable to choose bands solely based on the magnitude of ρ. This is because within a high-resolution continuous spectral range, the bands with the highest ρ values often exhibit high similarity, resulting in redundant data in the subset of bands. Fig. 2(a) presents a line plot of ρ values for each band in the dataset of the Pavia University, while Fig. 2(b) shows the 3-D visualization results after performing PCA, including all the bands. It can be observed that the top ten bands with the highest ρ values (marked by gray points) correspond to adjacent spectral wavelengths and are very close in the 3-D space. In fact, representative spectral bands should be widely distributed and contain rich spectral information within a local range. As shown in Fig. 2(a), the representative subset of spectral bands should consist of bands located at the peaks (marked by black points). Based on the black points in Fig. 2(b), we have summarized the characteristics of these representative spectral bands: they generally have ρ values higher than their neighbors; there is a relatively larger distance between spectral bands with higher ρ values. This observation is similar to cluster centers in density peak clustering [47], which prompted us to introduce an additional measure δ to assist in the exploration of representative bands, expressed as

$$\delta_i = \begin{cases} \min\left(D_{ij}\right), & \text{s.t } \rho_j > \rho_i \\ \max\left(D_{ij}\right), & \text{s.t } \rho_i = \max(\rho) \end{cases} \tag{10}$$

where D_{ij} denotes the Euclidean distance (ED) between bands $X(i,:)$ and $X(j,:)$, i.e.,

$$D_{ij} = \|X(i,:),\ X(j,:)\|_2. \tag{11}$$

By employing such calculations, we can ascertain that the band positioned at the peak of the "ρ value" exhibits significant δ. Consequently, we utilize the product of ρ and δ as a comprehensive metric to select representative bands, which is

$$\varphi_i = \rho_i \times \delta_i. \tag{12}$$

Once the computation of φ, we can effectively rank all bands in a descending order based on their φ values and subsequently select the top bands. This approach enables us to acquire a refined subset of bands that harmonizes the dual characteristics of high information content and low redundancy. As a result,

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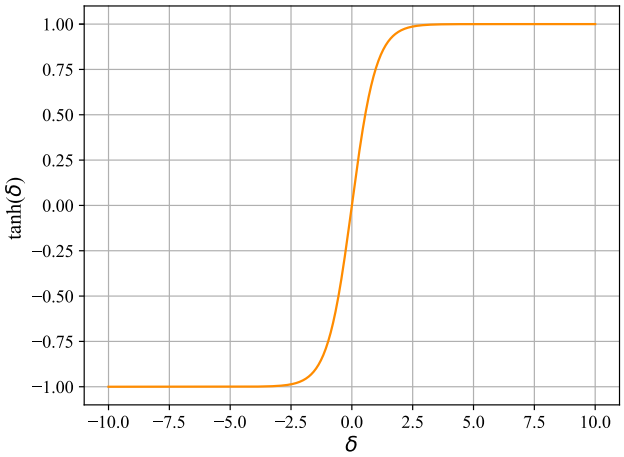


Fig. 3. Hyperbolic tangent curve.

the selected bands exhibit significant representation, aligning with the goals of our research.

In practical implementation, to mitigate the potential bias in band ranking caused by the magnitude disparity between ρ and δ, we perform standardization on these two metrics, mapping their ranges to the interval [0, 1], where ρ can be achieved by

$$\rho = (\rho - \rho_{\min})./(\rho_{\max} - \rho_{\min}) \tag{13}$$

where $\rho_{\max}$ and $\rho_{\min}$ are the minimum and maximum values of ρ. It is important to highlight that noisy bands caused by sensors and imaging systems have low SNR, rendering them inconsequential for target recognition. These noisy bands are discontinuous throughout the entire spectral range and typically exist as isolated points in the data space, consequently exhibiting a significant distance from the normal bands. Therefore, it is plausible that the conventional approach of linear normalization (13) may result in an excessively high δ value pertaining to the noisy band, consequently leading to an inflated subsequent ranking. To address this predicament, we have employed a nonlinear strategy to address the normalization of δ. Considering the saturation property of the hyperbolic tangent function, which can effectively reduce the impact of extreme values and enhance noise resistance [48], we have opted for the utilization of it as the transformation mechanism, as follows:

$$\delta = \tan \mathrm{h}(\delta) = \left(e^{\delta} - e^{-\delta}\right)./\left(e^{\delta} + e^{-\delta}\right) \tag{14}$$

graphically depicted in Fig. 3. Through this transformational process, the data is compressed to [0, 1]. Furthermore, it can be observed that the hyperbolic tangent function displays a decreasing growth rate in the first quadrant. Consequently, this transformation effectively reduces the amplification of "large distances," thereby effectively diminishing the likelihood of noisy bands being selected.

C. Fusion of Spatial–Spectral Information

In terms of band selection, the fusion of spatial context information plays a crucial role in identifying bands with rich structural features. It can effectively suppress noisy pixels and eliminate outliers. Particularly in this article, by considering the spatial relationships between adjacent pixels and applying the principle of smoothing, it is possible to reduce the interference of noisy pixels and outliers on the weighting coefficient ρ, thereby significantly improving the quality and reliability of the selected bands. However, traditional methods such as spatial filtering reduce the efficiency for the overall task when dealing with the aforementioned issues. In our research, we consider that image interpolation algorithms have the potential to address these problems. This is because the image interpolation scaling algorithms generate low-resolution images by weighted calculations of neighboring original pixels around the target pixel, thereby implicitly incorporating spatial context information. It is worth emphasizing that as the image is downscaled in spatial resolution, the overall computational time also decreases significantly. Bicubic interpolation, favored for its ability to effectively simulate the natural variations of images, is widely used in image processing [49]. To accurately preserve image structural features, we employed bicubic interpolation for scaling the original images (Fig. 1). Its mathematical formulation is detailed below

$$\check{X}(i+v, j+v) = \sum_{o=-1}^{2} \sum_{q=-1}^{2} X(i+o, j+q) \cdot \times w(o-v) w(q-u) \tag{15}$$

where $\check{X} \in \mathbb{R}^{l \times m}$ represents the interpolated image, where m is the downscaled image pixel count and $r = ((n/m))^{1/2}$ represents the spatial scaling factor. The bicubic convolution function, denoted as $w(\cdot)$, is used, and its formula is as

$$w(z) = \begin{cases} (a+2)|z|^3 - (a+3)|z|^2 + 1, \\ \qquad 0 \le |z| \le 1 \\ a|z|^3 - 5a|z|^2 + 8a|z| - 4a, \\ \qquad 1 < |z| \le 2 \\ 0 \\ \qquad \text{otherwise} \end{cases} \tag{16}$$

the parameter a is set to −0.5. Compared with other algorithms, the bicubic interpolation algorithm demonstrates its unique advantages in image scaling [50]. By considering the weights of 16 nearest neighboring pixels. The clever application of this strategy achieves an effective balance between the quality of band selection and computational efficiency.

The SIFDR workflow is outlined in Algorithm 1, which presents the pseudocode.

D. Complexity Analysis

Based on the derivation of pseudocode in Algorithm 1, the SIFDR algorithm involves key steps: bicubic interpolation for HSIs, inverse computation of the autocorrelation matrix R, and ρ calculation for band information estimation. Bicubic interpolation has a complexity of $O(ml)$, and Step 2.2 has complexities of $O(ml^2)$ and $O(l^3)$ for R and $R - 1$ on the downscaled image. The ρ calculation relies on the cost $O(ml^2)$ of W. The overall complexity is $O(2ml^2 + l^3 + ml)$. In Section IV-E, a sensitivity analysis determined that spatial

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Algorithm 1 DGWCR Workflow

INPUT: Raw HSIs $X \in \mathbb{R}^{l \times n}$, and the number of selected bands t.
OUTPUT: Selected Hypercube ***Step 1: Fusion of spatial information***
1: Generate the downscaled image $\check{X}$ using bicubic interpolation by *eq*.15
Step 2: Spectral information estimation
2: Calculate the autocorrelation matrix R of $\check{X}$
3: Construct weight matrix W by *eq*. 6
4: Calculate normalized bands' weights ρ by *eq*. 7 and 13
Step 3: Band weak redundancy ranking
5: Calculate distance matrix D using *eq*. 11
6: Calculate normalized bands' measures δ by *eq*. 10 and 14
7: Calculate comprehensive ranking metric φ by *eq*. 12
8: Rank the bands in descending order based on φ values
9: Select the top t bands from X to be $Y \in \mathbb{R}^{t \times n}$
10: **Return** Y

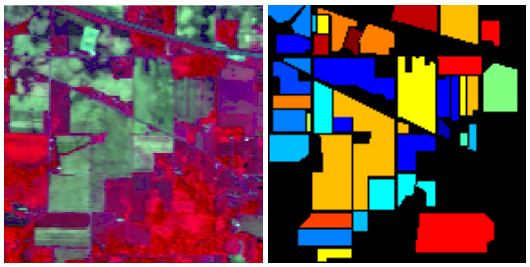


Fig. 4. (a) Pseudocolor image and (b) groundtruth map of IndianP.

scaling factor r is set to 8. Consequently, $m = (1/8^2)n$, the total time complexity of the SIFDR algorithm can be expressed as $O((1/32)nl^2 + l^3 + (1/64)nl)$, showcasing a substantial reduction in order of magnitude.

IV. Experiments

To assess the overall performance of the proposed SIFDR, comparative experiments are conducted using several SOTA methods on publicly available datasets. In contrast to previous studies, one medical HSIs is incorporated in our experiments, in addition to three remote-sensing datasets.

A. Datasets

1) Indian Pines (IndianP): The image was acquired by the airborne visible/infrared imaging spectrometer (AVIRIS) over the northwestern region of Indiana in 1992 [Fig. 4(a)]. It possesses a spatial dimension of 145 × 145 and a spatial resolution of 20 m. The image encompasses a compilation of 16 distinct land cover types [Fig. 4(b)], captured using a total of 220 spectral bands. Previous research often involves eliminating 20 bands due to potential noise resulting from water absorption or instrument artifacts. Conversely, in this study, we intentionally retained these bands to evaluate the effectiveness of the proposed method in managing noisy HSIs.

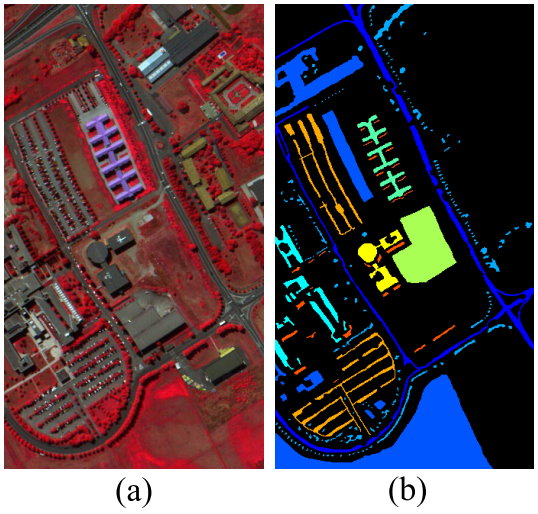


Fig. 5. (a) Pseudocolor image and (b) groundtruth map of PaviaU.

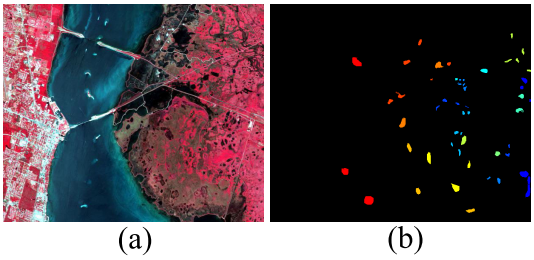


Fig. 6. (a) Pseudocolor image and (b) groundtruth map of KSC.

2) Pavia University (PaviaU): The data were collected at the University of Pavia in Italy using the reflective optics system imaging spectrometer system (ROSIS) within the wavelength range of 430–860 nm [Fig. 5(a)]. The ROSIS system can capture images of 103 spectral bands while maintaining a high spatial resolution of up to 1.3 m. The resulting image comprises 610 × 340 pixels, with certain pixels labeled under nine distinct land cover types [Fig. 5(b)].

3) Kennedy Space Center (KSC): The image in question, similar to IndianP, was acquired using the AVIRIS sensor and captured above the KSC in Florida, USA, in 1996 [Fig. 6(a)]. This image possesses a spatial resolution of 18 m and consists of 512 × 614 pixels. It exhibits 13 distinct categories [Fig. 6(b)]. Notably, it is capable of capturing 224 spectral bands ranging from 400 to 2500 nm with a 10 nm spatial resolution, and the public version has undergone the removal of noise bands caused by water absorption, resulting in the preservation of 176 spectral bands.

4) In ViVo Human Brain (Brain): The image is obtained from a database of live human brains that was published by the European project called hyperspectral imaging for cancer detection (HELICoiD) [51]. It is noteworthy that this database is one of the scarce publicly accessible repositories of medical HSIs. The medical HSIs were obtained using the Hyperspec VNIR A-series camera, manufactured by HeadWall Photonics Inc. (Fitchburg, MA, USA). The camera is capable of capturing up to 826 spectral bands, with a remarkable spatial resolution of 2–3 nm, encompassing wavelengths from 400 to

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TABLE I
DETAILS OF THE FOUR HSIS

Dataset	Spectral Bands	Spectral range	Total pixels	Classes	Spatial resolution
IndianP	220	400-2500nm	145 × 145	16	20m
PaviaU	103	430-860nm	610 × 340	9	1.3m
KSC	176	400-2500nm	512 × 614	13	18m
Brain	826	400-1000nm	443 × 497	4	128.7μm

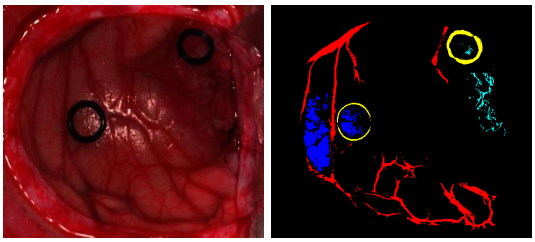


Fig. 7. (a) Pseudocolor image and (b) groundtruth map of brain.

1000 nm. In this study, we have chosen an image from a patient afflicted with grade IV glioblastoma (GBM) tumor [Fig. 7(a)]. The image depicts a scene comprising 443 × 497 pixels, offering a high spatial resolution (128.7 μm). Within this image, certain pixels have been precisely labeled to represent normal tissue, tumor tissue, blood vessels, or background elements [Fig. 7(b)]. To ensure precise analyses, calibration of the image's reflectance before usage is essential, following the methodology outlined by [52] to mitigate the impact of illumination and instrument noise.

The parameters of each image have been specifically summarized in Table I.

B. Compared Methods

To comprehensively evaluate the performance of the proposed method, a series of SOTA algorithms that have recently been proposed is compared with the SIFDR.

ASPS [29]: Through the utilization of adaptive subspace partitioning of the original image cube, the objective is to identify the band within the subcube that exhibits the least amount of noise.

FNGBS [30]: A strategy of spectral band grouping, using the concepts of coarse-fine, is employed. Subsequently, the most pertinent and informative bands are chosen from each group to form a subset.

ADBH [31]: A adaptive distance is employed to gauge the similarity between bands. Subsequently, hierarchical clustering is conducted on the bands, with local density serving as a priority in identifying representative bands.

RLLFC [27]: The superpixel is obtained through segmentation to extract the region-aware latent features of total bands. These shared latent feature representations of HSIs are then fused to cluster the bands.

HCBC [32]: Adaptive hyperbolic clustering is utilized to enhance the effective utilization of geometric information. The prioritization of band sorting is based on the potential degree of becoming a cluster center.

SGAE [53]: A novel random gate-based autoencoder is proposed for band selection. To improve the training process, a nonlinear regularization term loss is introduced, resulting in enhanced performance compared to the two existing deep learning band selection models, namely BS-Net [40] and CAE-Net [25].

HLFC [37]: Based on the segmentation of HSIs areas, a hierarchical strategy is employed to learn latent features, enhance the separability between bands through feature fusion, and utilize clustering to identify the band with the highest information entropy within each cluster.

MOCS [24]: The search capability of the cuckoo search algorithm is enhanced by implementing an information-sharing strategy that incorporates grouping and crossover. Moreover, to improve the stability of band selection, multiple targets of band subsets are optimized while considering the overall characteristics of HSIs.

IGAEBS [33]: The GCN trained on a pretext task optimizes the graph representation utilized to describe the structural information of HSIs. It clusters the bands based on the graph that is iteratively improved and introduces a new metric called IcSDD for band selection. The achieved results outperform those of DARecNet [41] and BS-Net.

C. Classification Context

The effectiveness of various methods for acquiring band subsets is evaluated by analyzing the classification results of three well-established classifiers: K-nearest neighbors (KNNs) [54], support vector machine (SVM) [55], and LDA [56]. The number of neighbors in KNN is set to three, while the penalty coefficient and gamma coefficient in SVM are set to optimized values obtained through tenfold cross-validation. In the experiments, 10%, 1%, and 10% of randomly selected labeled pixels are used for training in the IndianP, PaviaU, and KSC datasets, respectively. For the Brain dataset, 5 pixels are selected as training samples for each class, and the remaining labeled pixels are used for testing. The classification evaluation metrics used are overall accuracy (OA) and Kappa coefficient (KAPPA) [24]. To mitigate random errors, the average results of 20 independent runs are recorded. Most of the algorithms are implemented in MATLAB 2020a and executed on a computer with an Intel Core i9-11900 2.50 GHz CPU and 128GB of memory. Additionally, the two SOTA deep learning-based algorithms, GAE-UBS and IGAEBS, are implemented in Python 3.9 and accelerated using an NVIDIA Tesla A100 GPU (160 GB).

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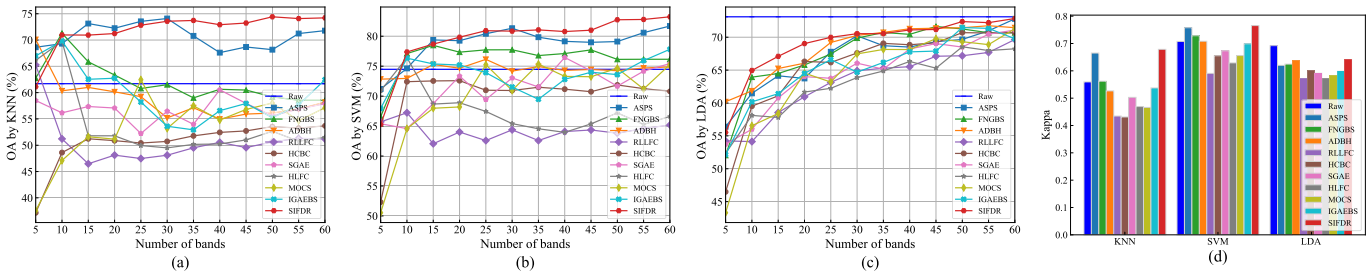


Fig. 8. OA of IndianP classification for different band selection methods from (a) KNN, (b) SVM, (c) LDA, and (d) Mean Kappa.

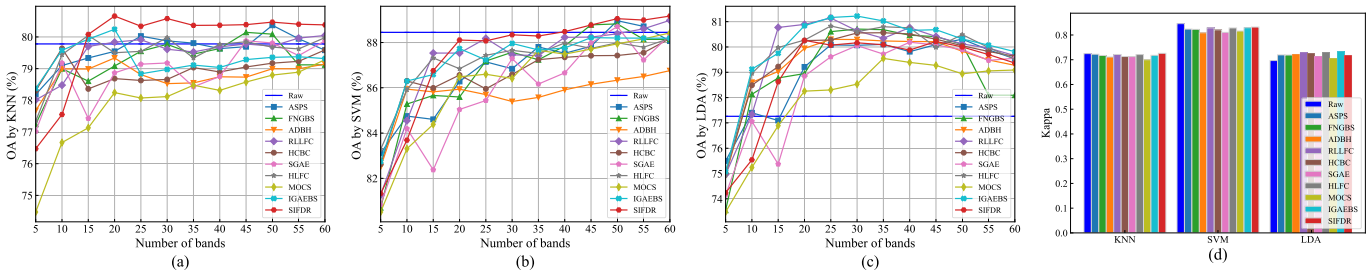


Fig. 9. OA of PaviaU classification for different band selection methods from (a) KNN, (b) SVM, (c) LDA, and (d) Mean Kappa.

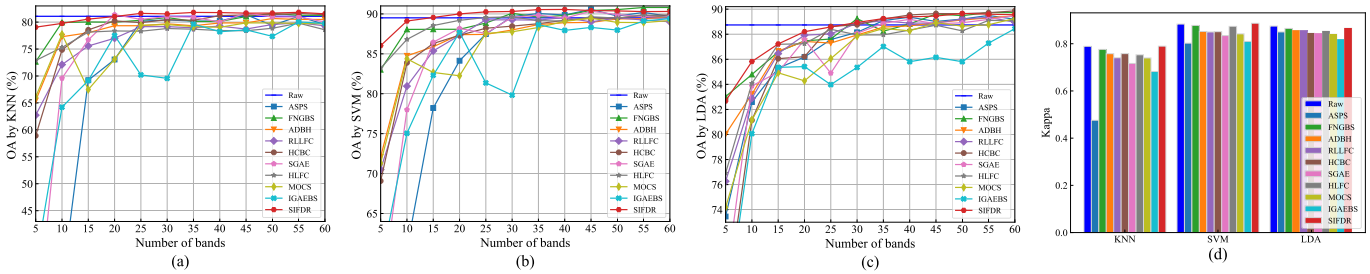


Fig. 10. OA of KSC classification for different band selection methods from (a) KNN, (b) SVM, (c) LDA, and (d) Mean Kappa.

D. Classification Results and Analysis

Figs. 8–11 depict the temporal classification accuracy of various algorithms for four HSIs with a selection of 5–60 bands (in increments of 5). The OA values are visually represented in line graphs Figs. 8(a)–(c)–11(a)–(c), while the bar chart Figs. 8(d)–11(d) records the average Kappa values associated with different numbers of bands.

In Fig. 8, it can be observed that for IndianP, the proposed SIFDR remains stable at a high level when faced with three classifiers. The classification accuracy generally improves with the increase of bands, particularly when performing KNN and SVM classification. In most cases, the accuracy of SIFDR is higher than that of the original data. Furthermore, Fig. 8(d) demonstrates that SIFDR achieves the highest kappa value. In contrast, some algorithms fail to exhibit stability when dealing with KNN classifiers. As OA reaches its peak, it decreases with the increase of bands. Moreover, MOCS and HCBC consistently perform at a low level, possibly due to the inclusion of intentionally retained noise bands. Overall, only ASPS demonstrates relative competitiveness, but the highest accuracy is still achieved by SIFDR.

Fig. 9 displays the classification accuracy of each algorithm on the PaviaU dataset. The results reveal that SIFDR performs better on KNN and SVM classifiers. Specifically, it consistently achieves higher than the OA of raw data after selecting 15 and 40 bands, surpassing other methods. Additionally, in the case of LDA classification, most algorithms exhibit similar levels of accuracy, with the exception of MOCS, which shows relatively lower performance.

Fig. 10 illustrates the accuracy of different algorithms on three classifiers for the KSC dataset. It can be clearly seen that SIFDR achieves better accuracy than other methods when selecting any number of bands. This point is better reflected by the average kappa value presented in Fig. 10(d). In relative terms, the classification accuracy of SIFDR on SVM is better, and in most cases, the OA of SIFDR exceeds 90%, which is higher than the accuracy of the original data. Notably, when considering Figs. 8–10 collectively, some algorithms perform well on one dataset but poorly on another, indicating their lack of universality. In contrast, the proposed SIFDR demonstrates stable performance.

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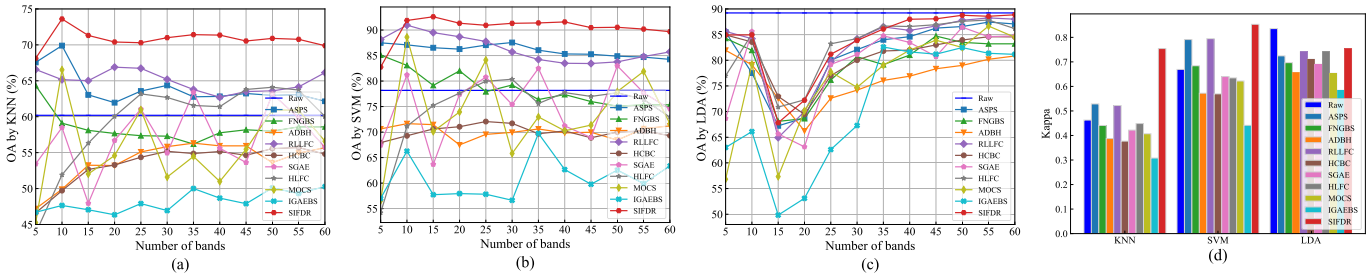


Fig. 11. OA of Brain classification for different band selection methods from (a) KNN, (b) SVM, (c) LDA, and (d) Mean Kappa.

TABLE II
CLASSIFICATION RESULTS OF DIFFERENT METHODS FOR FOUR HSIS

Datasets	Classifiers	Metrics	ASPS	FNGBS	ADBH	RLLFC	HCBC	SGAE	HLFC	MOCS	IGEBS	SIFDR
IP	KNN	OA(%)	67.07±0.60	60.64±0.66	54.96±0.71	49.90±0.73	52.44±0.45	60.53±0.65	50.92±0.52	51.95±0.63	56.57±0.66	**72.92±0.53**
		Kappa	0.6221±0.0069	0.5479±0.0075	0.4829±0.0083	0.4243±0.0077	0.4531±0.0050	0.5471±0.0075	0.4356±0.0056	0.4489±0.0067	0.5009±0.0072	**0.6897±0.0061**
	SVM	OA(%)	78.60±0.78	77.11±0.72	74.30±0.79	63.46±0.76	71.16±0.85	76.45±0.71	66.75±0.81	66.15±0.95	72.80±0.75	**80.78±0.66**
		Kappa	0.7562±0.0088	0.7392±0.0080	0.7069±0.0090	0.5829±0.0085	0.6711±0.0095	0.7316±0.0081	0.6203±0.0094	0.6141±0.0108	0.6900±0.0087	**0.7809±0.0075**
	LDA	OA(%)	68.90±0.60	70.44±0.73	**71.29±0.70**	66.10±0.55	68.85±1.00	68.22±0.81	65.85±0.72	65.37±0.57	67.77±0.78	71.11±0.64
		Kappa	0.6430±0.0072	0.6617±0.0083	**0.6716±0.0079**	0.6090±0.0063	0.6423±0.0116	0.6352±0.0095	0.6066±0.0083	0.6013±0.0068	0.6299±0.0090	0.6690±0.0075
PU	KNN	OA(%)	80.28±0.92	79.63±0.86	78.74±0.80	79.63±0.84	78.89±0.79	78.75±0.81	79.57±0.88	79.29±0.99	79.04±0.86	**80.37±0.96**
		Kappa	0.7316±0.0127	0.7229±0.0118	0.7104±0.0110	0.7229±0.0114	0.7126±0.0109	0.7106±0.0111	0.7219±0.0121	0.7187±0.0136	0.7144±0.0118	**0.7330±0.0132**
	SVM	OA	88.43±0.95	87.85±0.79	85.91±0.96	87.79±0.71	87.35±0.82	86.66±0.89	87.13±0.99	88.18±0.82	87.76±0.76	**88.48±0.80**
		Kappa	0.8458±0.0128	0.8381±0.0105	0.8122±0.0128	0.8372±0.0093	0.8315±0.0110	0.8223±0.0116	0.8284±0.0131	0.8426±0.0108	0.8368±0.0101	**0.8466±0.0108**
	LDA	OA(%)	80.46±0.74	80.46±0.76	80.22±0.68	80.49±0.67	80.51±0.76	80.19±0.71	80.09±0.72	79.18±0.65	**80.65±0.79**	79.88±0.69
		Kappa	0.7387±0.0099	0.7388±0.0101	0.7351±0.0091	0.7391±0.0089	0.7393±0.0104	0.7348±0.0095	0.7335±0.0098	0.7216±0.0085	**0.7412±0.0106**	0.7310±0.0091
KSC	KNN	OA(%)	79.91±0.84	80.11±0.83	79.84±0.74	79.19±0.83	81.20±0.66	80.00±0.79	78.48±0.82	73.93±0.86	78.21±0.80	**81.77±0.72**
		Kappa	0.7758±0.0094	0.7781±0.0093	0.7750±0.0083	0.7675±0.0093	0.7901±0.0074	0.7766±0.0089	0.7598±0.0091	0.7090±0.0095	0.7566±0.0089	**0.7965±0.0080**
	SVM	OA(%)	89.83±0.55	89.82±0.74	88.90±0.69	89.50±0.70	89.21±0.61	89.47±0.69	89.18±0.77	85.84±0.76	87.94±0.57	**90.56±0.71**
		Kappa	0.8867±0.0061	0.8865±0.0082	0.8763±0.0077	0.8830±0.0078	0.8797±0.0068	0.8827±0.0076	0.8795±0.0086	0.8422±0.0085	0.8656±0.0064	**0.8949±0.0079**
	LDA	OA(%)	88.82±0.49	88.82±0.47	88.66±0.55	89.00±0.68	**89.54±0.60**	89.19±0.44	87.90±0.76	87.62±0.65	85.81±0.63	89.39±0.51
		Kappa	0.8755±0.0054	0.8755±0.0052	0.8737±0.0062	0.8775±0.0076	**0.8835±0.0067**	0.8796±0.0049	0.8652±0.0084	0.8620±0.0072	0.8419±0.0071	0.8818±0.0057
Brain	KNN	OA(%)	63.95±6.06	57.76±4.20	55.95±4.59	63.11±6.22	55.16±4.86	55.61±4.21	62.64±4.23	53.76±3.68	48.67±2.85	**71.36±6.06**
		Kappa	0.5075±0.0791	0.4329±0.0562	0.4130±0.0499	0.5010±0.0771	0.3982±0.0523	0.3999±0.0576	0.4899±0.0535	0.3801±0.0426	0.3139±0.0434	**0.6030±0.0745**
	SVM	OA(%)	85.49±9.07	77.35±8.54	69.84±5.47	83.73±7.28	70.22±7.06	71.22±5.89	76.48±6.78	72.74±7.84	62.65±8.78	**91.59±6.24**
		Kappa	0.7830±0.1256	0.6689±0.1131	0.5693±0.0670	0.7570±0.0980	0.5718±0.0869	0.5853±0.0797	0.6580±0.0886	0.6053±0.1023	0.4707±0.1073	**0.8701±0.0880**
	LDA	OA(%)	84.57±5.40	81.00±6.71	76.90±7.31	84.06±5.33	81.99±6.83	83.46±6.66	85.87±7.06	81.98±5.87	81.63±7.85	**87.99±6.50**
		Kappa	0.7677±0.0733	0.7181±0.0894	0.6643±0.0919	0.7599±0.0716	0.7303±0.0913	0.7517±0.0868	0.7852±0.0981	0.7314±0.0787	0.7300±0.1015	**0.8161±0.0925**

Fig. 11 corresponds to the classification accuracy of the medical HSIs Brain dataset, and the source literature indicates that the Brain contains more than 100 noisy bands. Under these severe conditions, many algorithms began to fail, for instance, two recently proposed deep learning algorithms, SGAE and IGAEBS, showed poor performance. Only ASPS, RLLFC, and SIFDR consistently achieved higher accuracy than the original data in SVM and KNN, with SIFDR being the highest. In Fig. 11(c), all methods exhibit an accuracy trend of initially decreasing and then increasing. This may be due to a small number of noisy bands being involved in the classification initially, followed by the gradual emergence of more informative bands, only SIFDR approaches the classification accuracy of the original data when selecting 60 bands. Considering the KAPPA presented in Fig. 11(d), the proposed SIFDR demonstrates the best performance.

The classification results of three classifiers for different algorithms are recorded in Table II based on this. The best OA and Kappa values are highlighted in bold. It can be visually observed that the highest accuracy in both KNN and SVM classification for all datasets is achieved by SIFDR, with smaller standard deviations. This indicates that the stability of classification is enhanced by the bands obtained by SIFDR. Although the best accuracy on IndianP, PaviaU, and KSC datasets using ADBH, IGEBS, and HCBC, respectively, is achieved by LDA, the difference with SIFDR is negligible. It is worth noting that the best results across all classifiers for the Brain dataset are consistently achieved by SIFDR, demonstrating a significant advantage. The OA for KNN, SVM, and LDA are, respectively, 7.41%, 6.1%, and 2.12% higher than the second-ranked method, and an OA > 90% can only be achieved by SIFDR-SVM. Overall, more reliable subsets of bands for the four datasets are provided by the proposed SIFDR, demonstrating superior performance.

E. Parameter Sensitivity Analysis

This method involves a unique adjustable parameter r. To further assess the influence of r on SIFDR, we conduct an analysis of the classification outcomes by varying the values of r. Specifically, r is set to {1, 2, 4, 8, 16, 32}, and then the OA of SVM classification for 5–60 (in increments of 5) spectral bands is computed. To enhance the visual representation of the data, we presented the statistical results in a violin plot, as depicted in Fig. 12. The classification results on IndianP, PaviaU, and KSC datasets remain high and insensitive to changes when r ={2, 4, 8, 16}, as illustrated in Fig. 12. However, the classification effect appears to be unstable when $r = 1$ or 32. This instability is caused by the lack of spatial information when $r = 1$, highlighting

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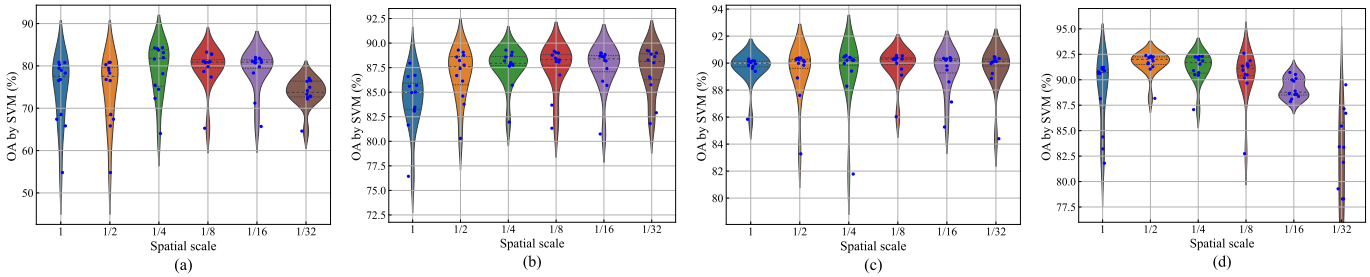


Fig. 12. Classification OA by SVM for HSIs. (a) IndianP, (b) PaviaU, (c) KSC, and (d) Brain at different r values.

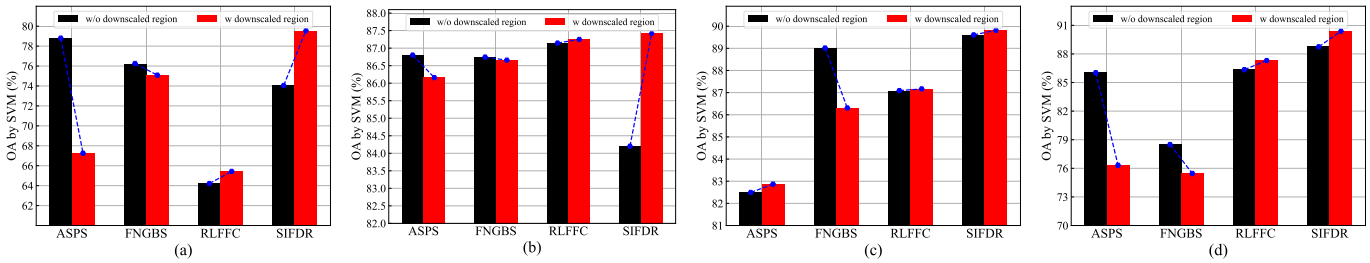


Fig. 13. Before and after image downscaling, OA of several excellent methods for four HSIs. (a) IndianP, (b) PaviaU, (c) KSC, and (d) SVM classification.

the need for spatial–spectral information fusion. Conversely, when $r = 32$, the image becomes too smooth, resulting in the loss of detailed information, which is more significant for medical images with higher spatial resolutions. As a result, the classification performance of Brain declines when $r = 16$, indicating the crucial role of an appropriate scale in effectively utilizing the spatial–spectral information of HSI. In summary, the intersection of r values that yield stable results across the four datasets is $\{2, 4, 8\}$. Considering the time performance, we fix $r = 8$ for SIFDR to approximately achieve nonparameterization.

F. Contribution of Spatial–Spectral Information Fusion

To validate the impact of spatial–spectral information fusion within the SIFDR framework, we performed ablation experiments. These experiments primarily involved comparing the classification performance before and after the downscaled region. The results captured the average classification accuracy of the SVM classifier across varying numbers of spectral bands for four datasets. To offer a thorough analysis, we additionally present the outcomes of the top-performing algorithms discussed in Section III-C, represented in Fig. 13.

In most scenarios, the classification performance of the ASPS and FNGBS methods deteriorates significantly in the downscaled region, as illustrated in Fig. 13. This is primarily because these methods primarily focus on band statistical information during their design, which not only fails to utilize spatial–spectral information based on the downsized image but also exacerbates the problem by losing some fine details. Conversely, in the RLLFC method, which extracts region-aware latent features, the classification accuracy is enhanced under downscaling, thereby demonstrating the efficacy of spatial–spectral information fusion. However, the improvements observed are only marginal, possibly due to the challenges in obtaining reasonable segmentation results (superpixels) for small-sized images in the presence of noisy bands. Compared with RLLFC, the proposed SIFDR framework exhibits a more substantial improvement (as indicated by the slope of the blue dashed line), achieving the optimal average OA for all datasets. This performance gain largely results from the ability of bicubic interpolation to mitigate noise speckles, thus enhancing band evaluation accuracy. Additionally, during downscaling, the concentration of noise within bands increases, leading to a higher likelihood of these bands being excluded during the weak redundancy ranking process. This unequivocally showcases the superior compatibility of the spatial–spectral fusion module within SIFDR.

G. Expression of Spatial–Spectral of the Selected Bands

In this section, we employ the more challenging medical HSIs Brain to assess the rationality of band subsets to express spatial–spectral information.

First, we take 40 bands selected by different methods to form a high-dimensional data space and reduce it to 2-D through t-SNE to visualize the distribution of samples from different categories (Fig. 14). It is apparent that in the raw HSIs, the presence of numerous noisy bands results in the mixture of samples from different categories, particularly normal tissue and blood vessels [Fig. 14(a)]. After band selection, most methods exhibit similar effects to the raw image. For instance, SGAE minimizes the reconstruction error of the raw HSIs, while RLFFC and HLFC introduce superpixels, both of which are affected by noise. In contrast, ASPS and the proposed SIFDR show good class separability and intraclass clustering, with SIFDR having almost no confusion pixels. Therefore, it fully demonstrates the evident spectral discrimination ability of the bands selected by SIFDR.

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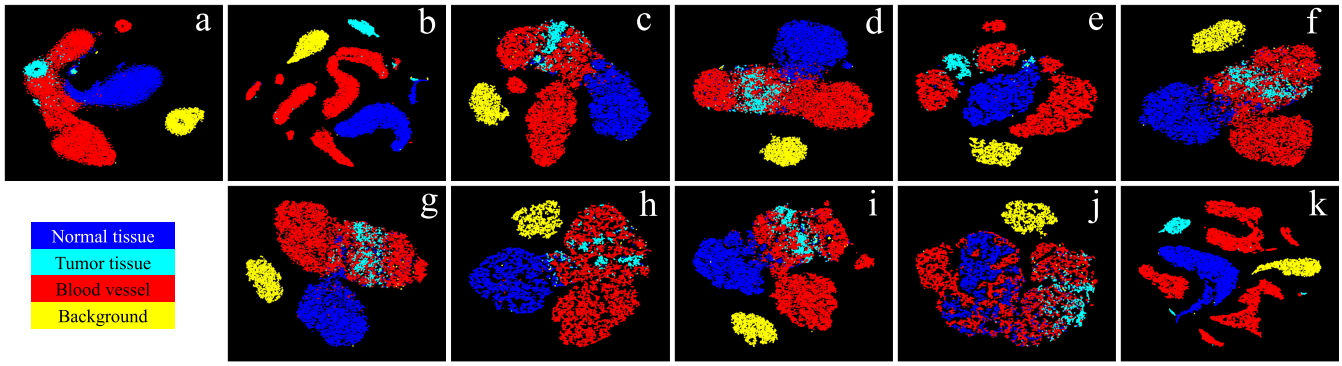


Fig. 14. Two-dimensional spatial distribution of labeled samples in (a) raw data or selected bands by (b) ASPS, (c) FNGBS, (d) ADBH, (e) RLLFC, (f) HCBC, (g) SGAE, (h) HLFC, (i) MOCS, (j) IGAEBS, and (k) SIFDR.

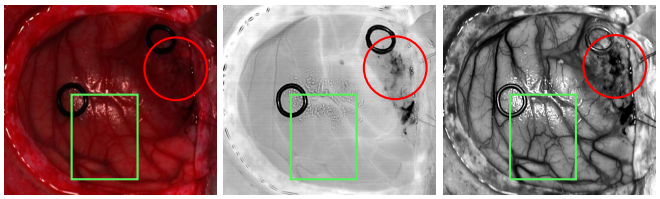


Fig. 15. Truecolor image. (a) Brain and the maps of the first principal component for (b) raw data and (c) selected hypercube by SIFDR.

TABLE III
TIME CONSUMING FOR SELECTING 40 BANDS WITH DIFFERENT ALGORITHMS (IN SECONDS)

Methods	IndianP	PaviaU	KSC	Brain
ASPS	0.074	0.281	0.498	2.398
FNGBS	0.855	3.726	10.847	102.361
ADBH	0.216	1.112	3.516	16.183
RLLFC	0.87	2.096	4.174	19.114
HCBC	0.21	1.042	3.066	13.98
SGAE	2.975	26.297	43.501	54.76
HLFC	3.872	4.078	6.923	38.791
MOCS	28.502	23.734	83.524	1113.709
IGAEBS	46.103	214.704	649.486	2553.356
SIFDR	**0.037**	**0.054**	**0.076**	**0.231**

Furthermore, we evaluated the spatial expressive performance of the selected bands by visualizing the first principal component of the new image cube generated by HSIs and SIFDR (Fig. 15). By observation, the first principal component map of SIFDR, clear textures were presented [Fig. 15(c)], revealing the details of capillaries within the green box and tumor tissues within the red circle. These details are not easily observable in the raw HSIs [Fig. 15(a)] and the conventional RGB image [Fig. 15(b)]. This observation unequivocally indicates the remarkable spatial expressive ability of SIFDR.

H. Efficiency Analysis

Table III displays the runtime of various algorithms on four datasets for the selection of 40 bands, with the shortest time highlighted in bold. It is evident that the proposed SIFDR algorithm accomplishes the task in an exceptionally short time and exhibits clear advantages compared to other methods. The runtime for the IndianP, PaviaU, and KSC datasets is all below 0.1 s, even on the Brain dataset with over 800 bands, it remains within a quarter of a second. This strongly indicates the real-time capability of the SIFDR algorithm, which is lacking in most existing algorithms.

V. CONCLUSION

A real-time unsupervised hyperspectral band selection framework, based on downscaled region for spatial–spectral information fusion, is proposed in this article. To estimate spectral information for each band, an average weight vector is generated in the spectral domain based on the constrained energy. Afterward, the bands are ranked with weak redundancy based on the weight peaks, while the introduction of hyperbolic tangent normalization suppresses noise, ultimately the complementarity of spectral information is enhanced. Additionally, we aim to achieve real-time processing by integrating spatial–spectral structural information into smaller sized images through a region fusion strategy. The proposed framework, SIFDR, is evaluated on traditional remote sensing HSIs and an ultrahigh-dimensional medical HSI. Comparative analysis with several SOTA band selection algorithms shows that SIFDR reduces data redundancy and demonstrates significant advantages in terms of classification performance and real-time capability. Nevertheless, the framework's universality remains untested on datasets with diverse sensors, scenarios, and spatial resolutions. Given the broad application of hyperspectral imaging in various fields, developing versatile band selection algorithms is crucial. Moving forward, our objective is to further validate and optimize the SIFDR algorithm to ensure its robustness in processing any HSIs.

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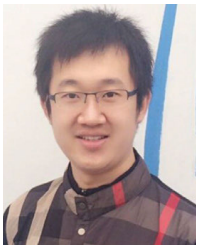
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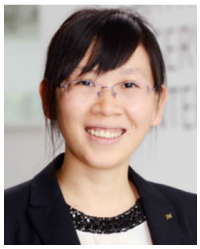
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